

Math 170S: Statistics

Winter 2019

Instructor:	Elizaveta (Liza) Rebrova
E-mail:	rebrova@math.ucla.edu (please, put "170S" in the subject line of your email)
CCLE Webpage:	https://ccle.ucla.edu/course/view/19W-MATH170S-1 (used for announcements only; please subscribe to receive them)
Course Webpage:	http://www.math.ucla.edu/~rebrova/170S.html (all crucial course info, syllabus and homework will be here)
Anonymous feedback form:	goo.gl/xNxFNb
Lectures:	
Room	6229 MS
Time	MWF 12:00am – 12:50am
Office Hours: (tentative)	
Room	5338 MS
Hours	M 2–3:30PM; F 10–11:30AM (or by appointment)
Teaching Assistant	John Zhang
Discussion session room	6229 MS
Discussion session hours	T 12:00am – 12:50am
Office hours	M 1-2 pm; T 11-12am

Course description

Lecture, three hours; discussion, one hour. Requisites: courses 31A, 31B, 170E. Introduction to statistics. Topics include sampling, estimation and properties of estimators, and construction of confidence intervals and hypotheses testing.

Course materials

- **Textbook:** Hogg, Tanis, Zimmerman Probability and Statistical Inference, 9th edition
- **Auxiliary textbook:** Introduction to Probability by D. P. Bertsekas and John N. Tsitsiklis, 2nd edition

Grading

Your grade is formed based on 8 homework assignments, one midterm, and the final exam. MyUCLA gradebook will be used for grading. The final score will be computed by the following scheme:

- homeworks - 15%, the midterm - 35%, the final - 50%

Any grading complaints and requests for re-grading must be submitted **within 1 week of receiving your score**. Grading complaints not initiated within this period of time will not be considered.

Homework

- Homework assignments will be issued weekly and posted on the course website <http://www.math.ucla.edu/~rebrova/170S.html>. It is turned in the beginning of each discussion section (and returned on the next discussion section).
- Your score on homework assignment will be based on a few randomly chosen problems, as well as overall completion of the assignment.
- **Late homework will not be accepted.**
- One lowest homework score will be dropped.
- Printed or handwritten homeworks have to be submitted.
- You may use whatever resources you want to do the homework, including computers, textbooks, friends, the TA, etc. Collaboration in small groups (up to 5 people) is encouraged. You can indicate the names of your collaborators if you'd like to. However, **all homework assignments must be written by you, i.e. you cannot copy someone else's solution or submit a joint solution made by several people.**

Exams

- There will be one midterm and one final exam.

Midterm: February 11 (Monday), in class

Final: March 21 (Thursday), 3–6pm, room TBA

- You **must take the final exam** in order to pass the class.
- No make-up exams will be possible (unless exceptional circumstances, validated by the Undergraduate office).
- Students must bring their UCLA ID cards to the midterms and to the final exam.

Tentative course schedule

This is a *very* tentative schedule:

Week	Date	Section	Topics
1-1	M, Jan 7	6.1	Descriptive Statistics
	T	Discussion	
1-2	W, Jan 9	6.2	Exploratory Data Analysis
1-3	F, Jan 11	6.3	Order Statistics
2-1	M, Jan 14	6.4	Maximum Likelihood Estimation
	T	Discussion	HW 1 collected
2-2	W, Jan 16	6.4	Maximum Likelihood Estimation
2-3	F, Jan 18	6.5	A Simple Regression Problem
3-1	M, Jan 21	No class	Martin Luther King, Jr, holiday
	T	Discussion	HW 2 collected
3-2	W, Jan 23	6.7	Sufficient Statistics
3-3	F, Jan 25	6.8	Bayesian Estimation
4-1	M, Jan 28	6.8	Bayesian Estimation
	T	Discussion	HW 3 collected
4-2	W, Jan 30	7.1	Confidence Interval for Means
4-3	F, Feb 1	7.2	Confidence Intervals for the Difference of Two Means
5-1	M, Feb 4	7.3	Confidence Interval for Proportions
	T	Discussion	HW 4 collected
5-2	W, Feb 6	7.4	Sample Size
5-3	F, Feb 8	7.5	Distribution-Free Confidence Intervals for Percentiles
6-1	M, Feb 11	6.1-7.5	Midterm
	T	Discussion	
6-2	W, Feb 13	8.1	Tests About One Mean
6-3	F, Feb 15	8.1	Tests About One Mean
7-1	M, Feb 18	No class	Presidents Day holiday
	T	Discussion	HW 5 collected
7-2	W, Feb 20	8.2	Tests of the Equality of Two means
7-3	F, Feb 22	8.3	Tests About Proportions
8-1	M, Feb 25	8.4	The Wilcoxon Tests
	T	Discussion	HW 6 collected
8-2	W, Feb 27	8.5	Power of a Statistical Test
8-3	F, Mar 1	8.6	Best Critical Regions
9-1	M, Mar 4	8.7	Likelihood Ratio Test
	T	Discussion	HW 7 collected
9-2	W, Mar 6	9.1	Chi-Square Goodness-of-Fit Tests
9-3	F, Mar 8	9.2	Contingency Tables
10-1	M, Mar 11	9.3	One-Factor Analysis of Variance
	T	Discussion	HW 8 collected
10-2	W, Mar 13	9.4	Two-Way Analysis of Variance
10-3	F, Mar 15	all	Review

Notice about academic integrity

From the office of the Dean of Students:

“With its status as a world-class research institution, it is critical that the University uphold the highest standards of integrity both inside and outside the classroom. As a student and member of the UCLA community, you are expected to demonstrate integrity in all of your academic endeavors. Accordingly, when accusations of academic dishonesty occur, The Office of the Dean of Students is charged with investigating and adjudicating suspected violations. Academic dishonesty includes, but is not limited to, cheating, fabrication, plagiarism, multiple submissions or facilitating academic misconduct.”

Students are expected to be aware of the University policy on academic integrity in the UCLA Student Conduct Code:

http://www.deanofstudents.ucla.edu/Portals/16/Documents/UCLACodeOfConduct_Rev030416.pdf

Please note the sections on (1) cheating, (2) plagiarism, and (3) unauthorized study aids.

Violation of course policy involving plagiarism, cheating, or possession of course materials during exams will be referred to the Dean of Students, who will be encouraged to take strong action. Do not cheat! The penalties can be very harsh. Do not believe it if you hear that “everyone does it.” You generally do not hear about the punishments because they are kept confidential. If you are found responsible by the Dean of Students for violating course policy, cheating on any course materials, or giving or receiving unauthorized help, a zero will be assigned for the entire assignment. No exceptions will be made! Past examples of penalties also include loss of an entire term of credit and suspension for several terms. If you plan to apply to graduate or professional school, such a negative mark on your record may be a major obstacle to admission.

No cell phones are allowed during exams. They must be left in your bag and turned off, or submitted to the designated TA/proctor. Students may not use a cell phone as a clock to keep time, nor as a calculator. No hats are allowed in the testing room; they must be left in your bag.

Notice about sexual harassment, discrimination, and assault

Title IX prohibits gender discrimination, including sexual harassment, domestic and dating violence, sexual assault, and stalking. Students who have experienced sexual harassment or sexual violence can receive **confidential** support and advocacy from a CARE advocate:

The CARE Advocacy Office for Sexual and Gender-Based Violence

1st Floor, Wooden Center West

CAREadvocate@caps.ucla.edu

(310) 206-2465

You can also report sexual violence or sexual harassment directly to the University’s Title IX Coordinator:

Kathleen Salvaty

2241 Murphy Hall

titleix@conet.ucla.edu

(310) 206-3417